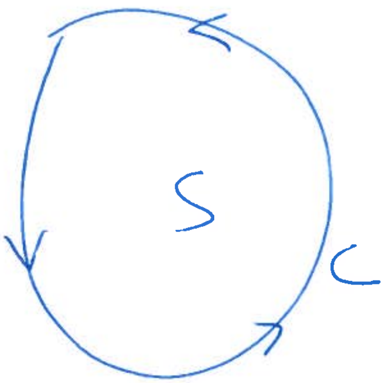
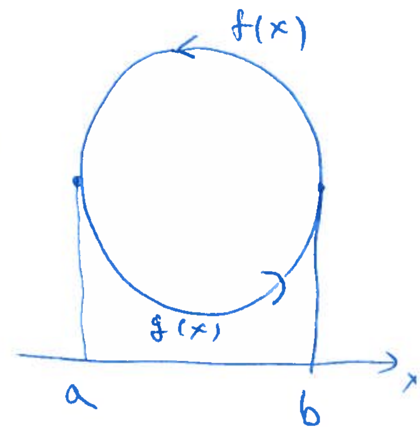
§ 14.4 Greens Theorem in the plane



C a smooth simple closed curve in the plane bounding a region S.

Then $\int_C M\,dx + N\,dy = \iint_S (N_x - M_y)\,dA$

Note: If $N_x = M_y$ then $F = M\,i + N\,j$ is conservativ & we know $\int_C M\,dx + N\,dy = 0$ frome 14.3.



Pf $\int_C M\,dx = \int_a^b M(x, g(x))\,dx + \int_a^b M(x, f(x))\,dx$

$= \int_a^b \Big(M(x, g(x)) - M(x, f(x))\Big)\,dx$

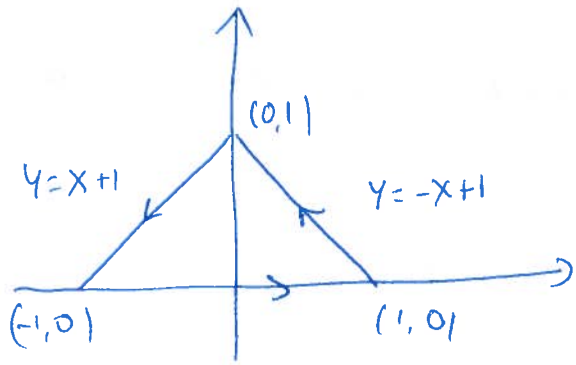
$= -\int_a^b \int_{g(x)}^{f(x)} \frac{\partial M(x,y)}{\partial y}\,dy\,dx = -\iint_S M_y\,dA$ $\square$

Eg C = circle of radius 2 center $(2,3)$

$\oint_C (2y + x^4)\,dx + (y^3 - x)\,dy = \iint_S +(y^3-x)_x - (2y+x^4)_y\,dA$

$= \iint_S -1 - 2\,dA = -3 \cdot \pi \cdot 2^2 = -12\pi$

Ex



$$\int_C (x^2+y^2)dx + xy\,dy$$

$$= \iint_S (xy)_x - (x^2+y^2)_y \; dA$$

$$= \int_0^1 \int_{y-1}^{1-y} y - 2y \quad dx\,dy$$

$$= \int_0^1 -y(1-y-y+1)\,dy = 2\int_0^1 y^2 - y\,dy = 2\left[\frac{y^3}{3} - \frac{y^2}{2}\right]_0^1 = 2\left[\frac{1}{3} - \frac{1}{2}\right] = -\frac{1}{3}$$

Ex $\frac{1}{2}\oint_C -y\,dx + x\,dy = \frac{1}{2}\iint 2\,dA = A(S)$

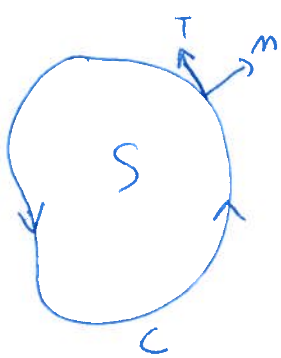
Ex Find area of ellipse $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$

$x = a\cos t$
$y = b\sin t$
$0 \le t \le 2\pi$

$$A(S) = \frac{1}{2}\int_C -y\,dx + x\,dy = \frac{1}{2}\int_0^{2\pi} -(b\sin t)(-a\sin t\,dt) + (a\cos t)(b\cos t\,dt)$$

$$= \frac{1}{2}\int_0^{2\pi} ab(\sin^2 t + \cos^2 t)\,dt = \frac{1}{2}ab\int_0^{2\pi} 1\,dt = \pi ab.$$

Ex

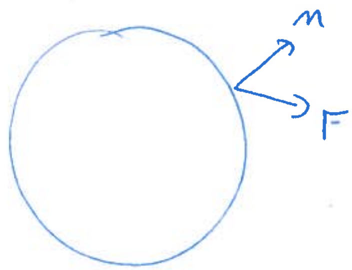


$$T = \left(\frac{dx}{ds}, \frac{dy}{ds}\right) \qquad n = \left(\frac{dy}{ds}, -\frac{dx}{ds}\right)$$

$$\int_C \underline{F}\cdot\underline{n}\,ds = \int (M,N)\cdot\left(\frac{dy}{ds}, \frac{-dx}{ds}\right)ds$$

$$= \int_C -N\,dx + M\,dy = \iint_S M_x + N_y\,dA$$

So $\int_C \underline{F}\cdot\underline{n}\,ds = \iint \operatorname{div} F\,dA$



$F \cdot n$ = component of F perp to C

$\int_C F \cdot n \, ds$ = flux of F across C

If div $F > 0$ we think of it as a source and if div $F < 0$ as a sink.

So the $\iint \operatorname{div} F \, dA$ represents the net influx of a "liquid" and it must be balanced by the flux across C.

Eg $F = \left(-\frac{1}{2}y, \frac{1}{2}x\right)$

$$\int_C F \cdot n \, ds = \iint_S \operatorname{div} F \, dA = \iint_S \left(-\tfrac{1}{2}y\right)_x + \left(\tfrac{1}{2}x\right)_y \, dA = 0$$

$$\int_C F \cdot T \, ds = \oint_C \left(-\tfrac{1}{2}y, \tfrac{1}{2}x\right) \cdot \left(\frac{dx}{ds}, \frac{dy}{ds}\right) ds = \int_C -\tfrac{1}{2}y \, dx + \tfrac{1}{2}x \, dy$$

$$= \iint_S \tfrac{1}{2} + \tfrac{1}{2} \, dA = A(S).$$

This also works in 3 dimensions

Gauss's Thm

$$\iint_{\partial S} F \cdot n \, dS = \iiint_S \operatorname{div} F \, dV$$

Ex $S = \{ x^2+y^2+z^2 \le 1 \}$ $\quad F = (x, y, z)$ $\quad n = (x, y, z)$

$\operatorname{div} F = 1+1+1 = 3$ $\qquad \iiint_S \operatorname{div} F = \frac{4}{3}\pi \cdot 3 = 4\pi$

$$\iint_{\partial S} F \cdot n \, dS = \iint_{\partial S} (x,y,z)\cdot(x,y,z)\, dS = \iint_{\partial S} x^2+y^2+z^2 \, dS = \iint_{\partial S} 1 \, dS$$

$$= 4\pi .$$

Ex S = tetrahedron with vertices $(0,0,0)$ $(1,0,0)$ $(0,1,0)$ $(0,0,1)$

$$\iint_{\partial S} (x+2yz, y+x^2, z-xy) \cdot n \, dS = \iiint_S 1+1+1 \, dV = 3 \cdot \operatorname{vol}(S)$$

$$= 3 \cdot \frac{1}{6} = \frac{1}{2}$$

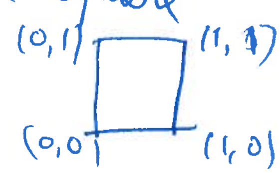
Ex Use Green's Thm to compute the following

a) $\int_C 2xy\,dx + y^2 dy$ where C is the close curve formed by $y = x/2$ & $y = \sqrt{x}$ between $(0,0)$ & $(4,2)$

b) $\int_C \sqrt{y}\,dx + \sqrt{x}\,dy$ where C is formed by $y=0, x=2$ & $y = \frac{x^2}{2}$

c) $\int_C (2x+y^2)dx + (x^2+2y)dy$ C is formed by $y=0$, $x=2$, $y = x^3/4$

Ex Use Green's Thm to compute $\int_C F \cdot n \, ds$ & $\int_C F \cdot T \, ds$ where a) $F = (y^2, x^2)$ C is the boundary of the square



b) $F = (y^3, x^3)$ C the unit circle

c) $F = (x, y)$ C the unit circle